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MACHINE LEARNING METHODS FOR CRYPTO LEVEL II DATA PREDICTION IN ALGORITHMIC TRADING

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ABSTRACT

In previous papers regarding market microstructure and price information in the stock market, the VAR model (vector autoregression) has been frequently mentioned as a benchmark, and there are various discussions about the pros and cons of this model. Here, we would like to extend the usage of this model to the cryptocurrency market and see how VAR performs in a new setting. Our result shows that although VAR might yield some gaps in terms of price prediction, it still catches the trend with a fair amount of accuracy and could still serve as a potential benchmark model.

Keywords: Algorithmic Trading, Machine Learning, Cryptocurrency, Bitcoin, Level II Order Book

1. INTRODUCTION AND DATA

The Crypto Level II order book provides detailed market data and information (S. Aleti and B. Mizrach, 2021) beyond the basic information available in a Level I order book (SN. Apergis, D. Koutmos, and J. E. Payne, 2021). For example, the Level II order book shows the depth of the market, revealing the quantity of buy and sell orders at different price levels. This includes not just the best bid and ask prices (as seen in Level I), but also additional orders lined up behind the best prices (SJ. Doering, M. Fairbank, and S. Markose, 2017). Traders can gauge market sentiment by analyzing the distribution and volume of buy and sell orders (J. Doyne Farmer 5, L. Gillemot, F. Lillo, S. Mike, and A. Sen, 2004). For instance, a large number of sell orders at a particular price level might indicate potential resistance (S. Haryanto, A. Subroto, and M. Ulpah, 2020), while a concentration of buy orders could signify support (M. Kearns and Y. Nevmyvaka, 2013).

We wrote a program in q/kdb+ to conduct data collection and processing. We concentrated on Bitcoin (sym: "X:SBTUSD@FTX") and selected 1st Aug, 2021 for our analysis. We then use the Q programming language to further create our four major features for our VAR model training. They are named as: "bid", "bsize", "ask" and "asize" respectively.

The "bid" variables refer to the top ten largest bid prices at a given timestamp, and the "bsize" variables represent the corresponding order sizes. The "ask" variable refers to the top 10 largest ask prices at a given timestamp, and the "asize" variables represent the corresponding order sizes. After suitable processing, we finally get a dataset with 159109 rows and 40 columns (features). The reason why we have 40 features is that we have 10 entries for each of the 4 major features and we decide to treat each one of them as an individual feature.

But to reduce the number of features, as proposed in (P. Lipinski and A. Brabazon, 2014), instead of directly using the top 10 bids, top 10 asks and their corresponding sizes as the features (T. Kuremoto, S. Kimura, K. Kobayashi, and M. Obayashi, 2014), what we will do is construct bid/ask blocks based on their relative distance (A. Kwan, R. Philip, and A. Shkillo, 2021) from the best bid and best ask.

First, we define something called a half-spread (h), which as the name suggests is half the bid-ask spread (G. Pieters, 2020).

Let B and A be the best bid and ask respectively (A. Yoshihara, K. Fujikawa, K. Seki, and K. Uehara, 2014). So we define the n th bid-block as including all bids whose values lie between $(B-(n-1)h, B-nh)$ (J. Puddick,

2017). Hence, the bid volume at block n is the sum of the bid sizes of all bids in this bid-block (W. Zhang and Y. Li, 2020). Similarly, we define the n th ask-block as well. n ranges from 1 to 9.

Thus, the best ask, best ask size, best bid and best bid size are 4 features. Then bid sizes of 9 bid blocks and ask sizes of 9 ask blocks form $9 \times 2 = 18$ more features, leading to a total of **22 features**. We will be using this for our analysis.

2. METHODS

VAR, or Vector Autoregression, is a statistical model used in econometrics and time series analysis. It's a type of model that captures the linear interdependencies among multiple time series.

Basic form of a VAR model:

A VAR model with p lags is written as:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_p Y_{t-p} + \varepsilon_t, \text{ where:}$$

Y_t is a vector of endogenous variables at time t ,

c is a constant vector (intercept),

$A_1, A_2, \dots, A_p$ are coefficient matrices,

ε_t is a vector of error terms.

One fascinating point of VAR is that it models multiple time series simultaneously. It's particularly useful when you want to understand and predict the behavior of several interrelated time series data, which is exactly what we are doing in this paper (the 22 features correlate with each other).

2.1. Preprocessing

As we have a total of 159,109 rows of data, we decided to split the whole dataset into 10 batches, each with a size of 15,000 (14980 for training and next 20 for testing). Also, for the VAR model to work, we must ensure that each of the 22 time-series is stationary. To check this, we run the ADF test (the `adfuller` function in python) on each of the series and perform differencing on the series that aren't stationary repeatedly until they are stationary. Our result shows that among the 22 variables, all features except for the best bid and the best ask are not stationary and they typically need one level of differencing.

2.2. Model Training and Prediction:

After the preprocessing process, we start to apply the VAR model (using `statsmodels.tsa.api.VAR()` function) on the data. We decided to train the VAR model on different batches of data (from batch 1 to batch 10), and then use the data at the subsequent 20 timestamps as test set to validate the quality of prediction. The metric we will be comparing between the forecast and actual data are bid prices, ask prices and mid prices, where mid prices can be simply calculated via the following equation:

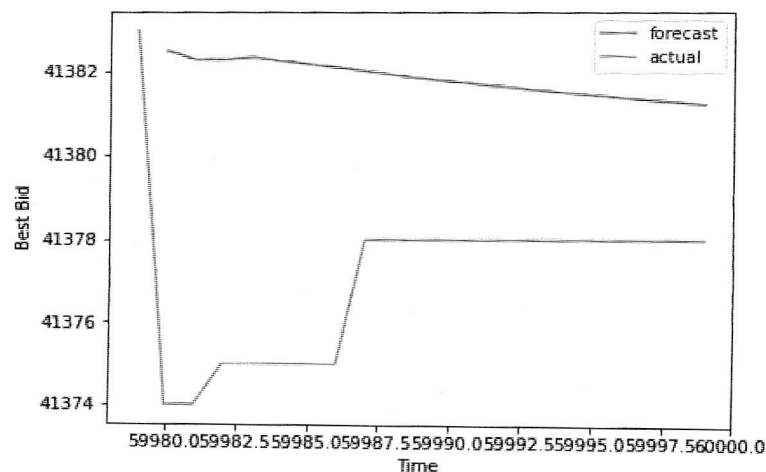
$$\text{mid price} = \frac{\text{bid price} + \text{ask price}}{2}$$

3. RESULTS

Let us see the result of a random batch, say 4th:

TABLE 1. TEST SET RESULT FOR THE VAR MODEL TRAINING ON THE FIRST BATCH OF DATA

	forecast			actual		
	bid_0	ask_0	mid_0	bid_0	ask_0	mid_0
59979	NaN	NaN	NaN	41383	41384	41383.5
59980	41382.500463	41384.164858	41383.332660	41374	41376	41375.0
59981	41382.287329	41383.926274	41383.106801	41374	41376	41375.0
59982	41382.287605	41383.965187	41383.126396	41375	41376	41375.5
59983	41382.338998	41383.838668	41383.088833	41375	41376	41375.5
59984	41382.261326	41383.735998	41382.998662	41375	41379	41377.0
59985	41382.184097	41383.599216	41382.891657	41375	41379	41377.0
59986	41382.114652	41383.527586	41382.821119	41375	41379	41377.0
59987	41382.021991	41383.455290	41382.738640	41378	41379	41378.5
59988	41381.931032	41383.356048	41382.643540	41378	41379	41378.5
59989	41381.838609	41383.276830	41382.557720	41378	41379	41378.5
59990	41381.773973	41383.187941	41382.480957	41378	41379	41378.5
59991	41381.703530	41383.107466	41382.405498	41378	41379	41378.5
59992	41381.638040	41383.031040	41382.334540	41378	41379	41378.5
59993	41381.569818	41382.959004	41382.264411	41378	41379	41378.5
59994	41381.506635	41382.889719	41382.198177	41378	41379	41378.5
59995	41381.446249	41382.825405	41382.135827	41378	41379	41378.5
59996	41381.387544	41382.763426	41382.075485	41378	41379	41378.5
59997	41381.332688	41382.703996	41382.018342	41378	41379	41378.5
59998	41381.279871	41382.646772	41381.963322	41378	41379	41378.5
59999	41381.229656	41382.592512	41381.911084	41378	41379	41378.5

FIGURE 1. FORECAST FOR THE BEST BID CAPTURES THE TREND IN THIS CASE

The result shows that our forecasts are not very close to the actual observed values but we do capture the trend:

1. The bid_value at the time of forecast was 41383 and at the end of forecast horizon was at 41378 (down by 5). And our forecast said it would be at 41381. So we did capture the direction right.
2. The ask_value at the time of forecast was 41384 and at the end of forecast horizon was at 41379 (down by 5). And our forecast said it would be at 41383. So here, we captured the direction right.
3. Same conclusion holds for mid-price as well.

Let's see if we have captured the trend for all the 10 batches:

TABLE 2. TEST SET RESULT FOR THE VAR MODEL TRAINING ON ALL THE 10 BATCHES OF DATA

	start			forecast		
	bid_0	ask_0	mid_0	bid_0	ask_0	mid_0
0	40932.0	40945.0	40938.5	40932.400063	40937.499285	40934.949674
15000	41434.0	41435.0	41434.5	41433.190197	41435.822440	41434.506319
30000	41437.0	41438.0	41437.5	41438.705378	41439.086526	41438.895952
45000	41383.0	41384.0	41383.5	41381.229656	41382.592512	41381.911084
60000	41500.0	41501.0	41500.5	41499.777422	41501.226524	41500.501973
75000	41318.0	41319.0	41318.5	41317.425607	41318.346311	41317.885959
90000	41188.0	41189.0	41188.5	41188.666305	41189.902761	41189.284533
105000	41229.0	41230.0	41229.5	41228.085737	41229.384592	41228.735165
120000	41074.0	41075.0	41074.5	41073.803086	41075.031028	41074.417057
135000	41004.0	41010.0	41007.0	41006.477356	41006.734683	41006.606020

	actual			results		
	bid_0	ask_0	mid_0	bid_0	ask_0	mid_0
0	40939.0	40940.0	40939.5	False	True	False
15000	41428.0	41429.0	41428.5	True	False	False
30000	41437.0	41438.0	41437.5	False	False	False
45000	41378.0	41379.0	41378.5	True	True	True
60000	41508.0	41509.0	41508.5	False	False	True
75000	41315.0	41316.0	41315.5	True	True	True
90000	41188.0	41189.0	41188.5	False	False	False
105000	41229.0	41230.0	41229.5	False	False	False
120000	41074.0	41075.0	41074.5	True	True	False
135000	41008.0	41009.0	41008.5	True	True	False
bid_0	5					
ask_0	5					
mid_0	3					

dtype: int64

The last set of columns in the summary table above tells us where all we predicted the trend right and where all we predicted wrong. So, as we see we predicted the bid correctly 5 times out of 10, ask 5 times out of 10 and mid 3 times out of 10. So, if we just consider the bid and the ask, this is 50% accuracy.

This is definitely better than a random guess. Because we have 3 possible outcomes - stay the same, go up and go down. Assuming all 3 are equally probable, a random guess would make us right 33% of the time. But we are achieving around 50%.

4. CONCLUSION

Although the VAR model prediction might generate gaps between the forecast and the actual value, it's still useful in terms of catching the price trend of bid prices, ask prices and mid prices in the high-frequency cryptocurrency market (with accuracy of around 50% in our case). This fact partly explains why VAR models were used in previous paper as benchmark model. In the future, other models that have previously been popular in the stock market (e.g. XGBoost, ARIMA and CNN) can also be extended and examined in the cryptocurrency market, which is much more volatile than the stock market. Further analyses also need be done regarding the market impact of the trading strategies derived from the model.

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